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Deriving the autocovariances of powers of Markov-switching GARCH models, with applications to statistical inference

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Abstract

A procedure is proposed for computing the autocovariances and the ARMA representations of the squares, and higher-order powers, of Markov-switching GARCH models. It is shown that many interesting subclasses of the general model can be discriminated in view of their autocovariance structures. Explicit derivation of the autocovariances allows for parameter estimation in the general model, via a GMM procedure. It can also be used to determine how many ARMA representations are needed to identify the Markov-switching GARCH parameters. A Monte Carlo study and an application to the Standard & Poor index are presented.
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1. Introduction

Since the seminal paper by Hamilton (1989), the use of Markov-switching models (MSM) has become increasingly popular in dynamic econometrics. A MSM is a discrete-time stochastic process $(\varepsilon_t, \Delta_t)$ such that: (a) (Δ_t) is a finite-state homogeneous Markov chain and, (b) the conditional distribution of ε_k, relative to its entire past, depends on (Δ_t) only through Δ_k. The process (Δ_t) is not observable and is often referred to as the *regime*. Flexibility is one of the main advantages of MSM. The changes in regime can be smooth or abrupt, and they can occur frequently or occasionally depending on the transition probabilities of the chain.

For financial time series, the generalized autoregressive conditional heteroskedasticity (GARCH) models, introduced by Engle (1982) and Bollerslev (1986), is certainly the most popular class of volatility models. Empirical works have often shown that a single standard GARCH model is hardly valid over a long period. For instance Baum and Karasulu (1998), and many references therein, have found evidence of regime changes for exchange rate data. This is the main motivation for considering the class of Markov-switching GARCH (MS-GARCH) models. The dynamics of the observed process ε_t is driven by a standard GARCH volatility as long as the chain Δ_t remains in the same regime, and is driven by a different standard GARCH volatility whenever the chain switches to another regime. Introducing changes in regime substantially increases the model flexibility, and allows interesting economic interpretations (the regimes of the Markov chain may correspond to economic phases, as in Hamilton and Lin, 1996). At the same time, the existence of

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several regimes entails serious complications. In particular the autocovariances of the squares of MS-GARCH models are hard to obtain due to the presence of a latent process in the volatility equation. Francq and Zakoïan (2005) (hereafter FZ) recently showed that, for all integer m, the autocovariance structure of (ε_t^{2m}) is that of an ARMA. That (ε_t^2) admits an ARMA representation is well known in the standard GARCH framework, but the result extends to higher powers and to the general MS case.

The primary purpose of this paper is to provide an operational procedure for computing the autocovariances and the ARMA representations of (ε_t^{2m}). We also investigate two statistical applications. First, we show that different subclasses of MS-GARCH models, namely the standard GARCH, the independent-switching GARCH, the hidden-Markov models (HMMs) and the Markov-switching ARCH, can be discriminated by considering their autocovariance structures. The subclasses imply different ARMA representations for ε_t^{2m} when the non-negative integer m varies. Considering only the ARMA representation for ε_t^2 will not suffice for statistical purposes. Specifically, the squares of a standard GARCH process and that of an independent-switching GARCH process may have the same ARMA representation, but higher powers will have different representations. Second, we implement a generalized method of moments (GMM) procedure (see Hansen, 1982) to estimate the MS-GARCH parameters. This is particularly relevant when certain regimes are not ARCH, because in this case, the usual estimation methods, such as the QMLE, are not available. Indeed, when all regimes are ARCH, algorithms, in the spirit of those developed by Hamilton (1989) (see e.g. Hamilton and Susmel, 1994; Cai, 1994; Hamilton and Lin, 1996) are available to compute the likelihood. However, such algorithms cannot be extended to MS-ARMA or MS-GARCH because conditionally to the chain, the process is not Markovian. Simulated methods, such as MCMC, have been advocated to approximate the likelihood in those types of models (see for instance the monograph by Robert and Casella, 2004 and references therein, and for recent references dealing with simulation-based estimation, see Carvalho and Lopes, 2007 in the framework of MS stochastic volatility models, and Giannikis et al., 2007 for a class of threshold normal mixture GARCH models). The GMM estimators proposed in the present paper are an alternative to these simulated methods which are computationally much more demanding. Our simulations suggest that these estimators behave well for moderate sample sizes. We illustrate the proposed techniques through an application to the daily series of the Standard & Poor 500. The evidence is quite strong that the series is not the realization of a standard GARCH process. A two-regime GARCH(1,1) model is shown to provide a reasonable fit to these data.

The article is organized as follows. Section 2 introduces the model and the main assumptions. Section 3 summarizes and illustrates results given in FZ concerning the autocovariance structure of (ε_t^{2m}). Section 4 provides a procedure to compute the autocovariances and to obtain the coefficients of the ARMA representations of (ε_t^{2m}). All our results are illustrated by the simplest non-trivial model, that is the 2-regime GARCH(1,1). Section 5 presents the statistical applications. Numerical illustrations are given in Section 6. Section 7 provides an empirical application to the daily series of the Standard & Poor 500. Section 8 concludes. A technical proof is given in the appendix.

2. MS-GARCH models

The Markov-switching GARCH(p, q) model that we consider here is given by

$$\begin{cases} \varepsilon_t = \sqrt{h_t}\eta_t, \\ h_t = \omega(\Delta_t) + \sum_{i=1}^{q} \alpha_i(\Delta_t)\varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j(\Delta_t) h_{t-j}. \end{cases} \tag{1}$$

The following assumption will be maintained throughout the paper.

Assumption A. (η_t) and (Δ_t) are independent processes; (η_t) is a sequence of independent and identically distributed (iid) random variables with zero mean and unit variance; (Δ_t) is an homogeneous, stationary, irreducible, and aperiodic Markov chain with finite state-space $\mathscr{E}=\{1, 2, \ldots, d\}$, stationary distribution $\pi(k) := P(\Delta_1=k)$, transition probabilities $p(k, \ell) := P(\Delta_t = \ell | \Delta_{t-1} = k)$, and n-step transition probabilities $p^{(n)}(k, \ell) = P(\Delta_t = \ell | \Delta_{t-n} = k)$ for $k, \ell \in \mathscr{E}$ and $n \geqslant 1$; $\omega(k) > 0$, $\alpha_i(k) \geqslant 0$, and $\beta_j(k) \geqslant 0$ for $k \in \mathscr{E}$, $i \in \{1, \ldots, q\}$, $j \in \{1, \ldots, p\}$.

Note that other MS-GARCH specifications have been proposed in the literature (see in particular Gray, 1996 or Haas et al., 2004). The MS-GARCH(p, q) specification (1) includes, as special cases, several classes of models that have

been investigated separately in the literature:

(i) *Standard* GARCH(p, q): This model is obtained by assuming the functions $\omega(\cdot)$, $\alpha_i(\cdot)$ and $\beta_j(\cdot)$ constant, or equivalently by assuming that the chain (Δ_t) has a single regime. The most widely used model of the empirical finance literature is the standard GARCH(1,1).
(ii) *Independent-switching* GARCH(p, q): In this specification, analyzed by Wong and Li (2001) in the ARCH case, (Δ_t) is an independent process. See Fong et al. (2006) for a multivariate extension.
(iii) *Hidden-Markov model* (HMM): In contrast with MSM, HMM introduced in statistics and engineering are characterized by the fact that given (Δ_t), (ε_t) is a sequence of independent random variables. This class is obtained by setting $h_t = \omega(\Delta_t)$, i.e. $\alpha_i(\cdot) = \beta_j(\cdot) = 0$, for all i and j in (1). Since the seminal paper by Baum and Petrie (1966), a growing statistical literature has been devoted to HMM (see Guédon, 2007 for a recent reference).
(iv) *Markov-switching* ARCH(q): This model, obtained for $p = 0$, is one of the most popular MS volatility models (see e.g. Hamilton and Susmel, 1994; Cai, 1994; Hamilton and Lin, 1996; Francq et al., 2001).
(v) *Stochastic volatility model* (GARCH(p, 0)): This special case is obtained by taking $\alpha_i(\cdot) = 0$ for all i. Contrary to the standard GARCH volatility, h_t remains non-deterministic when the ARCH coefficients vanish. It should be noted that this specification makes (h_t) and (η_t) independent and can therefore be viewed as a stochastic volatility model (see Shephard, 1996 for a survey of ARCH and stochastic volatility modeling).

Each of the previous models is a potential candidate to model the volatility of financial time series. Unfortunately, faced with the problem of choosing a particular specification among (i)–(v), empirical researchers do not have prior statistical tools at their disposal (such as, for example, the autocorrelation function for selecting an ARMA model, or the autocorrelation of the squares for standard GARCH specifications). The computation of the autocovariances of the squares and higher-order powers of the general model (1) will prove useful to this aim.

3. Autocovariance structure of (ϵ_t^{2m})

In this section we summarize and illustrate results given in FZ concerning the autocovariance structure of (ε_t^{2m}), where $m > 0$ be an integer such that $E\eta_t^{2m} < \infty$.

3.1. Existence of moments

Conditions for the existence of a stationary solution of (1) satisfying $E\varepsilon_t^m < \infty$ are presented in Table 1. These conditions involve the following notations. For any function $f : \mathscr{E} \to \mathscr{M}_{n\times n'}(\mathbb{R})$, where $\mathscr{M}_{n\times n'}(\mathbb{R})$ denotes the space of real $n \times n'$ matrices, and any positive integers i, n and n', let the matrices

$$\mathbb{P}^{(i)}(f) = \begin{pmatrix} p^{(i)}(1,1)f(1) & \cdots & p^{(i)}(d,1)f(1) \\ \vdots & & \vdots \\ p^{(i)}(1,d)f(d) & \cdots & p^{(i)}(d,d)f(d) \end{pmatrix} \quad \text{and} \quad \Pi(f) = \begin{pmatrix} \pi(1)f(1) \\ \vdots \\ \pi(d)f(d) \end{pmatrix}. \tag{2}$$

Table 1
Conditions for the existence of $E(\varepsilon_t^{2m})$ in the MS-GARCH(p, q) model (1)

Specification	General case	Case $p = q = 1$
Standard[a]	$\rho\{E_\eta(C_t^{\otimes m})\} < 1$	$E_\eta\{\alpha_1\eta_t^2 + \beta_1\}^m < 1$
IS[b]	$\rho\{E_{\eta,\Delta}(C_t^{\otimes m})\} < 1$	$E_{\eta,\Delta}\{\alpha_1(\Delta_t)\eta_t^2 + \beta_1(\Delta_t)\}^m < 1$
MS-ARCH[c]	$\rho\{\mathbb{P}(E_\eta(C_t^{\otimes m}\|\Delta_t = \cdot))\} < 1$	$E_\eta(\eta_t^{2m})\rho\{\mathbb{P}(\alpha_1^m(\cdot))\} < 1$
MS[d]	$\rho\{\mathbb{P}(E_\eta(C_t^{\otimes m}\|\Delta_t = \cdot))\} < 1$	$\rho\{\mathbb{P}(E_\eta\{\alpha_1(\cdot)\eta_t^2 + \beta_1(\cdot)\}^m)\} < 1$

[a]Standard: standard GARCH or MS-GARCH when only ω is switching.
[b]Independent-switching: $p(i, j)$ independent of i, for $i, j \in \mathscr{E}$.
[c]Markov-switching ARCH: $\beta_j(\cdot) = 0, \quad j = 1, \ldots, p$.
[d]Markov-switching: general model.

When $i=1$ the notation is simplified as $\mathbb{P}(f)=\mathbb{P}^{(1)}(f)$, and when $f\equiv 1$, the notation $\mathbb{P}(f)$ reduces to the transpose of the transition matrix $\mathbb{P}=(p(j,i))$. For $p>0$ and $q>0$, let the vectors $\alpha_{1:q-1}(\Delta_t)=(\alpha_1(\Delta_t),\ldots,\alpha_{q-1}(\Delta_t))$ and $\beta_{1:p-1}(\Delta_t)=(\beta_1(\Delta_t),\ldots,\beta_{p-1}(\Delta_t))$, and let the $(p+q)\times(p+q)$ matrix

$$C_t=\begin{pmatrix}\alpha_{1:q-1}(\Delta_t)\eta_t^2 & \alpha_q(\Delta_t)\eta_t^2 & \beta_{1:p-1}(\Delta_t)\eta_t^2 & \beta_p(\Delta_t)\eta_t^2\\ I_{q-1} & 0 & 0 & 0\\ \alpha_{1:q-1}(\Delta_t) & \alpha_q(\Delta_t) & \beta_{1:p-1}(\Delta_t) & \beta_p(\Delta_t)\\ 0 & 0 & I_{p-1} & 0\end{pmatrix},$$

where I_k denotes the $n\times n$ identity matrix. The definition of C_t can be modified in an obvious way when $p=0$ or $q=0$. Denote by $\rho(A)=\max_i\{|\lambda_i|;\ \lambda_i \text{ eigenvalue of } A\}$ the spectral radius of any square matrix A. Let $\otimes$ denote the Kronecker product defined for any matrices A, B by $A\otimes B=(a_{ij}B)$. For any matrix A, let $A^{\otimes m}=A\otimes\cdots\otimes A$. To avoid misunderstandings, in Table 1 we have indexed the expectations by the processes they involve.

Example 1. Consider the case of a MS-GARCH(1,1) with two regimes. FZ showed that, in this case, the condition given in Table 1 is also necessary for the existence of $E\varepsilon_t^{2m}$. Applying this condition with $m=1$, we find that $E\varepsilon_t^2$ is finite if and only if the spectral radius of

$$\begin{pmatrix}p(1,1)\{\alpha(1)+\beta(1)\} & p(2,1)\{\alpha(1)+\beta(1)\}\\ p(1,2)\{\alpha(2)+\beta(2)\} & p(2,2)\{\alpha(2)+\beta(2)\}\end{pmatrix}$$

is strictly less than one.

3.2. ARMA representations of (ε_t^{2m})

Under the assumption that $E\varepsilon_t^{4m}<\infty$ it is shown in FZ that the autocovariance structure of (ε_t^{2m}) is that of an ARMA. At least for the case $m=1$, this result is well-known for standard GARCH processes, but is much harder to show for $m>1$ in the general case of MS-GARCH models.

In the sequel we limit ourself to MS-GARCH(1,1) models, but allow arbitrary values of $m\geqslant 1$. Additional notations are needed. The $2m$-th order moment of the iid process is denoted by $\mu_{2m}=E(\eta_t^{2m})$. For $0\leqslant m''\leqslant m'\leqslant m$, let

$$\xi_{m',m''}(\cdot)=\binom{m'}{m''}\{\omega(\cdot)\}^{m'-m''}E[\alpha(\cdot)\eta_{t-1}^2+\beta(\cdot)]^{m''}\frac{\mu_{2m'}}{\mu_{2m''}}. \tag{3}$$

For any square matrix M, $n_0(M)$ denotes the algebraic multiplicity of the eigenvalue 0, and $n_1(M)$ denotes the sum of the algebraic multiplicities of the eigenvalues of module 1. Let the $d(m+1)\times d(m+1)$ matrix

$$A_m=\begin{pmatrix}\mathbb{P}(\xi_{m,m}) & \mathbb{P}(\xi_{m,m-1}) & \cdots & \cdots & \mathbb{P}(\xi_{m,0})\\ 0 & \mathbb{P}(\xi_{m-1,m-1}) & \cdots & \cdots & \mathbb{P}(\xi_{m-1,0})\\ \vdots & \ddots & \ddots & & \vdots\\ 0 & \cdots & 0 & \mathbb{P}(\xi_{1,1}) & \mathbb{P}(\xi_{1,0})\\ 0 & \cdots & 0 & 0 & \mathbb{P}\end{pmatrix}. \tag{4}$$

If $\alpha(\cdot)+\beta(\cdot)>0$, and $\rho\{\mathbb{P}(\xi_{2m,2m})\}<1$, then (ε_t^{2m}) is solution of an ARMA(N_1,N_2) equation, with orders $N_1=(m+1)\{d-n_0(\mathbb{P})\}-n_1(A_m)$ and $N_2=(m+1)\{d-n_0(\mathbb{P})\}-n_1(A_m)$.

Table 2 summarizes the available results concerning the orders and coefficients (when an analytic expression is available) of the ARMA representations, for different powers of MS-GARCH(1,1) processes. In this table, an ARMA representation is called strong if its noise is a martingale difference, and is called weak when its noise is just an uncorrelated sequence. For instance, it is known that if ε_t follows a standard GARCH(p,q), then ε_t^2 follows an ARMA$(\max\{p,q\},p)$ in which the noise is a martingale difference (thus a strong ARMA$(\max\{p,q\},p)$). For $m>1$, ε_t^{2m} still follows an ARMA, but the linear innovations of ε_t^{2m} is not a martingale difference sequence. Thus ε_t^{2m} is said to follow a weak ARMA representation. The distinction is important because the conditional expectation of ε_t^{2m} is linear (and thus the optimal predictor is given by the ARMA representation) if and only if the representation is strong.

Table 2
Characteristics of the ARMA representations for ε_t^{2m} in (MS-)GARCH(1, 1) models

Specification	Representation	AR polynomial
$m = 1$		
Standard[a]	Strong ARMA(1,1) (AR(1) iff $\beta_1 = 0$)	$1 - (\alpha_1 + \beta_1)L$ (MA polynomial $= 1 - \beta_1 L$)
IS[b]	Strong ARMA(1,1) (AR(1) iff $\beta_1(\cdot) = 0$)	$1 - E\{\alpha_1(\Delta_t) + \beta_1(\Delta_t)\}L$ (MA polynomial $\neq 1 - E\{\beta_1(\Delta_t)\}L$)
HMM[c]	ARMA($d-1, d-1$)	$\prod_{k=1}^{d-1}(1-\lambda_k L)$, $\lambda_k =$ eigenvalues ($\neq 1$) of $\mathbb{P}$
MS-ARCH[d]	ARMA($2d-1, 2d-2$)	$\prod_{k=1}^{2d-1}(1-\lambda_k L)$, $\lambda_k =$ eigenvalues ($\neq 1$) of $\mathbb{P}$ and $\mathbb{P}(\alpha_1)$
MS[e]	ARMA($2d-1, 2d-1$)	$\prod_{k=1}^{2d-1}(1-\lambda_k L)$, $\lambda_k =$ eigenvalues ($\neq 1$) of $\mathbb{P}$ and $\mathbb{P}(\alpha_1 + \beta_1)$
$m > 1$		
Standard	ARMA(m, m) (ARMA($m-1, m-1$) if $\beta_1 = 0$)	$\prod_{k=1}^{m}\{1 - E(\alpha_1\eta_t^2 + \beta_1)^k L\}$
IS	ARMA(m, m) (AR($m, m-1$) if $\beta_1(\cdot) = 0$)	$\prod_{k=1}^{m}\{1 - E(\alpha_1(\Delta_t)\eta_t^2 + \beta_1(\Delta_t))^k L\}$
HMM	ARMA($d-1, d-1$)	$\prod_{k=1}^{d-1}(1-\lambda_k L)$, $\lambda_k =$ eigenvalues ($\neq 1$) of $\mathbb{P}$
MS-ARCH	ARMA($md+d-1, md+d-2$)	$\prod_{k=1}^{md+d-1}(1-\lambda_k L)$, $\lambda_k =$ eigenvalues ($\neq 1$) of the $\mathbb{P}\{\alpha_1(\cdot)^i\}$, $i = 0, \ldots, m$
MS	ARMA($md+d-1, md+d-1$)	$\prod_{k=1}^{md+d-1}(1-\lambda_k L)$, $\lambda_k =$ eigenvalues ($\neq 1$) of the $\mathbb{P}[E\{\alpha_1(\cdot)\eta_t^2 + \beta_1(\cdot)\}^i]$, $i = 0, \ldots, m$

[a]Standard: single-regime GARCH(1,1).
[b]Independent-switching: $p(i, j)$ independent of i, for $i, j \in \mathscr{E}$.
[c]Hidden-Markov: $\alpha_1(\cdot) = \beta_1(\cdot) = 0$.
[d]Markov-switching ARCH: $\beta_1(\cdot) = 0$.
[e]Markov-switching: general model.

Example 2. Consider a MS-GARCH(1,1) model with two regimes. We have $\xi_{1,1} = \alpha(\cdot) + \beta(\cdot)$, $\xi_{1,0} = \omega(\cdot)$ and

$$A_2 = \begin{pmatrix} p(1,1)\{\alpha(1)+\beta(1)\} & p(2,1)\{\alpha(1)+\beta(1)\} & p(1,1)\omega(1) & p(2,1)\omega(1) \\ p(1,2)\{\alpha(2)+\beta(2)\} & p(2,2)\{\alpha(2)+\beta(2)\} & p(1,2)\omega(2) & p(2,2)\omega(2) \\ 0 & 0 & p(1,1) & p(2,1) \\ 0 & 0 & p(1,2) & p(2,2) \end{pmatrix}.$$

Now take for instance $p(1,1) = p(2,1) = 0.2$, $\alpha(1) = \beta(1) = 0.05$, $\alpha(2) = \beta(2) = 0.1$. Then the eigenvalues of A_2 are 0, 0, 0.18 and 1. Thus $n_0(A_2) = 2$, $n_1(A_2) = 1$ and (ε_t^2) is solution of an ARMA(1,1) equation with AR polynomial $1 - 0.18L$, where L denotes the lag operator. Note that, in this example, the actual orders $p = q = 1$ are smaller than the maximal orders $p = q = 2d - 1 = 3$ given in Table 2.

It suffices to compute the eigenvalues of $\mathbb{P}(\xi_{2,2})$ to obtain the AR polynomial of the representation for ε_t^4. Assuming that η_t is gaussian, we have $\xi_{2,2}(1) = 0.015$ and $\xi_{2,2}(2) = 0.06$. The non-zero eigenvalue of $\mathbb{P}(\xi_{2,2})$ is 0.051. We therefore obtain an ARMA(2,2) representation for ε_t^4 with AR polynomial $(1 - 0.18L)(1 - 0.051L)$.

4. Deriving the autocovariances and ARMA representations of ε_t^2 and its powers

The statistical applications developed further do not only require the existence of ARMA representations for ε_t^{2m}, but also require their explicit computation. For a given MS-GARCH(p,q) model, we show in this section how to compute the ARMA coefficients, as well as the autocovariance function of (ε_t^{2m}). For simplicity, we will only consider the GARCH(1,1) case. For convenience we recall the model

$$\begin{cases} \varepsilon_t = \sqrt{h_t}\eta_t, \\ h_t = \omega(\Delta_t) + \alpha(\Delta_t)\varepsilon_{t-1}^2 + \beta(\Delta_t)h_{t-1}, \end{cases} \tag{5}$$

and we omit the subscripts of the coefficients α and β. Under the assumption $E\varepsilon_t^{4m} < \infty$, we first compute the mean and autocovariances of ε_t^{2m}.

4.1. Deriving the autocovariances

Using the notation (2), define

$$W_m(\ell) = \begin{pmatrix} \Pi(E[\varepsilon_t^{2m}\varepsilon_{t-\ell}^{2m}|\Delta_t = \cdot]) \\ \Pi(E[\varepsilon_t^{2(m-1)}\varepsilon_{t-\ell}^{2m}|\Delta_t = \cdot]) \\ \vdots \\ \Pi(E[\varepsilon_{t-\ell}^{2m}|\Delta_t = \cdot]) \end{pmatrix}. \tag{6}$$

For $0 \leqslant m'' \leqslant m' \leqslant m$, let

$$\xi^*_{m,m',m''}(k) = \binom{m'}{m''} \{\omega(k)\}^{m'-m''} E[\{\alpha(k)\eta_{t-1}^2 + \beta(k)\}^{m''}\eta_{t-1}^{2m}] \frac{1}{\mu_{2(m''+m)}}. \tag{7}$$

The following result, which is proved in the appendix, allows to compute the autocovariances of ε_t^{2m} conditional on Δ_t.

Theorem 3. *Let* (ε_t) *be a solution of model* (5) *and assume*

$$\rho[\mathbb{P}(E\{\alpha(\cdot)\eta_t^2 + \beta(\cdot)\}^{2m})] < 1.$$

Then (ε_t^{2m}) *is second-order stationary and, using the notation* (3),

$$\Pi(E[\varepsilon_t^{2m'}|\Delta_t = \cdot]) = \begin{cases} \Pi(1) & \text{for } m' = 0 \\ \{I - \mathbb{P}(\xi_{m',m'})\}^{-1} \sum_{m''=0}^{m'-1} \mathbb{P}(\xi_{m',m''})\Pi(E[\varepsilon_t^{2m''}|\Delta_t = \cdot]) & \text{for } m' = 1, \ldots, 2m. \end{cases} \tag{8}$$

For $m' = 1, \ldots, m$

$$\Pi(E[\varepsilon_t^{2m'}\varepsilon_{t-1}^{2m}|\Delta_t = \cdot]) = \sum_{m''=0}^{m'} \mathbb{P}(\xi^*_{m,m',m''})\Pi(E[\varepsilon_t^{2(m''+m)}|\Delta_t = \cdot]). \tag{9}$$

For $\ell > 1$,

$$W_m(\ell) = A_m W_m(\ell - 1) \quad \text{where } A_m \text{ is defined by } (4). \tag{10}$$

This theorem allows to compute $W_m(\ell)$ for all $\ell \geqslant 0$. The unconditional means follow from $E(\varepsilon_t^{2m'}) = e_d'\Pi(E[\varepsilon_t^{2m'}|\Delta_t = \cdot])$, where $e_d = (1, \ldots, 1)'$ denotes the d element sum vector. The unconditional autocovariances similarly follow by summing appropriate components of $W_m(\ell)$.

Example 4. Consider the MS-GARCH(1,1) model with $d = 2$ regimes, transition probabilities $p(1,1) = p(2,1) = 0.2$, and GARCH regimes $\omega(1) = 1$, $\alpha(1) = \beta(1) = 0.05$, and $\omega(2) = 2$, $\alpha(2) = \beta(2) = 0.1$. Simple computations show

that $\Pi(1) = (\pi(1), \pi(2))' = (0.2, 0.8)'$. Applying (8) yields

$$\Pi(E[\varepsilon_t^2|\Delta_t = \cdot]) = \begin{pmatrix} \pi(1)E[\varepsilon_t^2|\Delta_t = 1] \\ \pi(2)E[\varepsilon_t^2|\Delta_t = 2] \end{pmatrix} = \begin{pmatrix} \dfrac{0.2}{0.82} \\ \dfrac{1.6}{0.82} \end{pmatrix},$$

$$\Pi(E[\varepsilon_t^4|\Delta_t = \cdot]) = \begin{pmatrix} 0.909 \\ 14.557 \end{pmatrix}.$$

Therefore, $E(\varepsilon_t^2) = 2.195$ and $E(\varepsilon_t^4) = 15.467$. Hence $\gamma_{\varepsilon^2}(0) = 10.648$. Now, applying (9) with $m = m' = 1$, we get

$$\Pi(E[\varepsilon_t^2\varepsilon_{t-1}^2|\Delta_t = \cdot]) = \begin{pmatrix} \pi(1)E[\varepsilon_t^2\varepsilon_{t-1}^2|\Delta_t = 1] \\ \pi(2)E[\varepsilon_t^2\varepsilon_{t-1}^2|\Delta_t = 2] \end{pmatrix} = \begin{pmatrix} 0.645 \\ 5.162 \end{pmatrix},$$

hence $\gamma_{\varepsilon^2}(1) = 0.989$. In view of the AR coefficient, and because ε_t^2 admits an ARMA(1,1) representation in this particular case (see Example 1), we have that $\gamma_{\varepsilon^2}(\ell) = (0.18)^{\ell-1}0.989$, for $\ell \geqslant 1$. To conclude, let us derive the MA coefficient, b say, as a solution to the second-order equation

$$\frac{b}{1+b^2} = \frac{\operatorname{cov}(\varepsilon_t^2 - 0.18\varepsilon_{t-1}^2, \varepsilon_{t-1}^2 - 0.18\varepsilon_{t-2}^2)}{\operatorname{var}(\varepsilon_t^2 - 0.18\varepsilon_{t-1}^2)} = -0.087.$$

Choosing the solution of module less than 1, we obtain the ARMA(1,1) representation:

$$\varepsilon_t^2 - 2.195 = 0.18(\varepsilon_{t-1}^2 - 2.195) + u_t - 0.088u_{t-1}, \quad \operatorname{var}(u_t) = 10.556.$$

In general, when $d > 2$, the ARMA representations involve higher orders and such computations cannot be done by hand. An algorithm is proposed in the next section to obtain the ARMA coefficients.

4.2. Deriving the ARMA representation

From Table 2 and FZ we know that, provided (ε_t^{2m}) is second-order stationary, it admits an ARMA representation of the form

$$\left(1 + \sum_{i=1}^{md+d-1} a_i^{(m)} L^i\right)\{\varepsilon_t^{2m} - E\varepsilon_t^{2m}\} = \left(1 + \sum_{i=1}^{md+d-1} b_i^{(m)} L^i\right) u_t^{(m)}. \tag{11}$$

It can be assumed, without generality loss, that (11) is the canonical ARMA representation, i.e. the AR and MA polynomials have their roots outside the unit circle and have no common root. In this section, we provide an algorithm for computing the coefficients $a_i^{(m)}$, $b_i^{(m)}$ and $\operatorname{var}(u_t^{(m)})$ in function of Model (5) parameters (i.e. $\omega(i)$, $\alpha(i)$, $\beta(i)$, $p(i, j)$, for $i, j \in \mathscr{E}$, and $\mu_{2m'} = E(\eta_t^{2m'})$, for $m' = 1, \ldots, 2m$). In the Gaussian case we have $\mu_{2m'} = (2m')!/(2^{m'}m'!)$. A practical approach to deriving the ARMA representation (11) proceeds as follows.

Step 1 (definition of intermediate tools): Define the functions

$$\xi_{m',m''}(k) = \binom{m'}{m''}\{\omega(k)\}^{m'-m''}\frac{\mu_{2m'}}{\mu_{2m''}}\sum_{m'''=0}^{m''}\binom{m''}{m'''}\alpha(k)^{m'''}\mu_{2m'''}\beta(k)^{m''-m'''},$$

$$\xi^*_{m,m',m''}(k) = \binom{m'}{m''}\{\omega(k)\}^{m'-m''}\frac{1}{\mu_{2(m''+m)}}\sum_{m'''=0}^{m''}\binom{m''}{m'''}\alpha(k)^{m'''}\mu_{2(m'''+m)}\beta(k)^{m''-m'''},$$

and $\mathbb{P}(\cdot)$, introduced in (3), (7), and (2), respectively.

Step 2 (moments existence): If $\rho(\mathbb{P}(\xi_{2m,2m})) \geqslant 1$, then terminate. Otherwise, go to Step 3.

Step 3 (computation of the autocovariances): Compute, in this order,

- $\Pi(E[\varepsilon_t^{2m'}|\Delta_t = \cdot])$ for $m' = 1, \ldots, 2m$, using (8).
- $\Pi(E[\varepsilon_t^{2m'}\varepsilon_{t-1}^{2m}|\Delta_t = \cdot])$ for $m' = 0, \ldots, m$ using (9).

- $\Pi(E[\varepsilon_t^{2m'}\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot])$ for $m'=0,\dots,m$ and $\ell=2,\dots,2m(d+1)-2$ using (10).
- $E\varepsilon_t^{2m}=e_d'\Pi(E[\varepsilon_t^{2m}|\Delta_t=\cdot])$, $\gamma_{\varepsilon^{2m}}(0)=e_d'\Pi(E[\varepsilon_t^{4m}|\Delta_t=\cdot])-(E\varepsilon_t^{2m})^2$,
- $\gamma_{\varepsilon^{2m}}(\ell)=e_d'\Pi(E[\varepsilon_t^{2m}\varepsilon_{t-\ell}^{2m}|\Delta_t=\cdot])-(E\varepsilon_t^{2m})^2$ for $\ell=1,\dots,2d(m+1)-2$.

Step 4 (computation of the AR coefficients): Compute the eigenvalues of $\mathbb{P}$, $\mathbb{P}(\xi_{1,1}),\dots,\mathbb{P}(\xi_{m,m})$ and replace those of module 1 by 0. Denote by $\lambda_1,\dots,\lambda_{md+d-1},0$ these numbers. Compute the $a_i^{(m)}$ from the recursion

$$\begin{cases} a_{0,0}=1, \\ a_{0,j}=1, \quad a_{i,j}=a_{i,j-1}-\lambda_j a_{i-1,j-1} \quad \forall i\in\{1,\dots,j-1\}, \quad a_{j,j}=-\lambda_j a_{j-1,j-1} \\ \quad \text{for } j=1,\dots,md+d-1, \\ a_i^{(m)}=a_{i,md+d-1} \quad \forall i\in\{1,\dots,md+d-1\}. \end{cases}$$

Step 5 (computation of the autocovariances of the AR part): Compute recursively

$$\begin{cases} \gamma^{(0)}(h)=\gamma_{\varepsilon^{2m}}(h) \quad \text{for } h=0,\dots,2(md+d-1), \\ \gamma^{(i)}(h)=\gamma^{(i-1)}(h)(1+\lambda_i^2)-\lambda_i\{\gamma^{(i-1)}(|h-1|)+\gamma^{(i-1)}(h+1)\} \\ \quad \text{for } i=1,\dots,md+d-1 \quad \text{and} \quad h=0,\dots,2(md+d-1)-i, \\ \gamma_{\mathrm{AR}}(h)=\gamma^{(md+d-1)}(h) \quad \text{for} \quad h=0,\dots,md+d-1. \end{cases}$$

and set $\gamma_{\mathrm{AR}}(h)=0,\ \forall h>md+d-1$.

Step 6 (computation of the MA coefficients and noise variance):

(1) Set $i=1$.
(2) Compute $b_{i,1},\dots,b_{i,i}$ and v_i, from the recursion relations

$$\begin{cases} v_0=\gamma_{\mathrm{AR}}(0), \\ b_{i,i-k}=v_k^{-1}\left\{\gamma_{\mathrm{AR}}(i-k)-\sum_{j=0}^{k-1}b_{i,i-j}b_{k,k-j}v_j\right\}, \quad k=0,\dots,i-1, \\ v_i=\gamma_{\mathrm{AR}}(0)-\sum_{j=0}^{i-1}b_{i,i-j}^2v_j. \end{cases}$$

(3) If $i>md+d-1$ and $\max_{k=1,\dots,md+d-1}\{|b_{i,k}-b_{i-1,k}|\}$ is less than a given positive tolerance then set $b_k^{(m)}=b_{i,k}$ for $k=1,\dots,md+d-1$. Otherwise, replace i by $i+1$ and go to 2.

A few comments are in order. In Step 2,

$$\rho(\mathbb{P}(\xi_{2m,2m}))=\rho[\mathbb{P}(E\{\alpha(\cdot)\eta_t^2+\beta(\cdot)\}^{2m})]<1$$

is a necessary and sufficient condition for the existence of an ARMA representation for ε_t^{2m}. In Step 4, note that at least one eigenvalue is equal to 1. The computation of the $a_i^{(m)}$ is derived from $1+\sum_{i=1}^{md+d-1}a_i^{(m)}L^i=\prod_{i=1}^{md+d-1}(1-\lambda_iL)$. In Step 5 we compute the autocovariances $\gamma_{\mathrm{AR}}(h)$ of the MA process

$$AR_t:=\left(1+\sum_{i=1}^{md+d-1}a_i^{(m)}L^i\right)\{\varepsilon_t^{2m}-E\varepsilon_t^{2m}\}=\prod_{i=1}^{md+d-1}(1-\lambda_iL)\{\varepsilon_t^{2m}-E\varepsilon_t^{2m}\}.$$

Simple computations show that $\gamma_{\mathrm{AR}}(h)=\prod_{i=1}^{md+d-1}(1-\lambda_iL)(1-\lambda_iL^{-1})\gamma_{\varepsilon^{2m}}(h)$, for all $h\in\mathbb{Z}$ with the convention that $L^k\gamma_{\varepsilon^{2m}}(h)=\gamma_{\varepsilon^{2m}}(h-k)$ for all $k\in\mathbb{Z}$. An alternative way to compute these autocovariances is to use

$$\gamma_{\mathrm{AR}}(h)=\sum_{i,j=0}^{md+d-1}a_i^{(m)}a_j^{(m)}\gamma_{\varepsilon^{2m}}(|h+j-i|) \quad \text{for } h=0,\dots,md+d-1.$$

In Step 6, we use the innovations algorithm presented in Brockwell and Davis (pp. 172, 245, 1991). By their Proposition 5.2.2 and Remark 3, p. 176, $\lim_{i\to\infty}v_i=\mathrm{Var}(u_t^{(m)})$, $\lim_{i\to\infty}b_{i,k}=b_k^{(m)}$, $\forall k\geqslant 1$, with $b_k^{(m)}=0$ for $k>md+d-1$.

Remark 5. In the particular case where $d=2$, the AR coefficients of Step 4 can be analytically obtained, thus avoiding numerical computation. The eigenvalues of $\mathbb{P}(\xi_{0,0})=\mathbb{P}$ are 1 and $\lambda_1 := 1-p(1,1)-p(2,2)$. Straightforward calculations show that, for $m\geqslant 1$, the sum of the eigenvalues of $\mathbb{P}(\xi_{m,m})$ is $s_m := p(1,1)\xi_{m,m}(1)+p(2,2)\xi_{m,m}(2)$, and their product is $p_m := \xi_{m,m}(1)\xi_{m,m}(2)(p(1,1)+p(2,2)-1)$. Hence, for $m>1$, the AR polynomial of ε_t^{2m} is obtained by multiplying the AR polynomial of $\varepsilon_t^{2(m-1)}$ by $1-s_mL+p_mL^2$. For instance, the AR coefficients $a_i^{(1)}$, $i=0,\ldots,3$ for $m=1$, and $a_i^{(2)}$, $i=0,\ldots,5$ for $m=2$ are

$$
\begin{aligned}
&a_0^{(1)}=1, \quad a_1^{(1)}=-s_1-\lambda_1, \quad a_2^{(1)}=p_1+s_1\lambda_1, \quad a_3^{(1)}=-p_1\lambda_1,\\
&a_0^{(2)}=1, \quad a_1^{(2)}=a_1^{(1)}-s_2, \quad a_2^{(2)}=a_2^{(1)}-s_2a_1^{(1)}+p_2, \quad a_3^{(2)}=a_3^{(1)}-s_2a_2^{(1)}+p_2a_1^{(1)},\\
&a_4^{(2)}=-s_2a_3^{(1)}+p_2a_2^{(1)}, \quad a_5^{(2)}=p_2a_3^{(1)}
\end{aligned}
$$

with λ_1 replaced by 0 if it is equal to 1 or -1.

Example 6. Consider the two-regime GARCH(1,1) defined by $\varepsilon_t=\sqrt{h_t}\eta_t$ with

$$h_t=\{1+0.05\varepsilon_{t-1}^2+0.05h_{t-1}\}1_{\{\Delta_t=1\}}+\{2+0.1\varepsilon_{t-1}^2+0.1h_{t-1}\}1_{\{\Delta_t=2\}},$$

where ε_t is iid $\mathcal{N}(0,1)$, $p(1,1)=p(2,1)=0.2$. Implementing the algorithm given in Steps 1–6 by means of a Fortran program, we obtain the following ARMA representations:

$$
\begin{aligned}
&(1-0.18L)(\varepsilon_t^2-2.195)=(1-0.088L)u_t^{(1)},\\
&(1-0.231L+0.00918L^2)(\varepsilon_t^4-15.467)=(1-0.1486L+0.0034L^2)u_t^{(2)}
\end{aligned}
$$

and

$$
\begin{aligned}
&(1-0.2541L+0.0145L^2-0.0002L^3)(\varepsilon_t^6-191.59)\\
&\quad=(1-0.1947L+0.0068L^2-0.00006L^3)u_t^{(3)},
\end{aligned}
$$

where the $u_t^{(i)}$ are white noises such that $\mathrm{var}(u_t^{(1)})=10.556$, $\mathrm{var}(u_t^{(2)})=3232.58$ and $\mathrm{var}(u_t^{(3)})=2795698$.

5. Statistical applications

This section proposes two statistical applications. The first subsection shows that the autocovariance structure results can be useful to identify a particular subclass of model (1). The second subsection uses the algorithm presented in Section 4.2 to estimate MS-GARCH models.

5.1. Identifying a class of MSMs

In many economic applications, it is of interest to know if the dynamics displays different regimes and which models are adequate for these regimes. In the framework of this paper, we are able to characterize the following five situations in terms of their ARMA representations: (a) standard (or single-regime) GARCH(p,q), (b) independent-switching GARCH(p,q), (c) HMM ($h_t=\omega(\Delta_t)$), (d) Markov-switching ARCH(q), and the general Markov-switching GARCH(p,q). Without building a formal identification procedure, which we consider a topic for future work, it is worth noting that the ARMA representations can be helpful to assess the relevance of the general model and its subclasses (a)–(d) for a given set of observations. Table 2 shows that the representations corresponding to $m=1$ are potentially useful for distinguishing models (a) and (b) from the other ones: the existence of a (minimal) ARMA(p_0,q_0) representation for ε_t^2 with $p_0>1$ or $q_0>1$, rules out the possibility of the first two specifications (and also that of a two-regime HMM). Conversely, if an ARMA(1,1) is identified for ε_t^2, a standard GARCH(1,1), an independent-switching GARCH(1,1) or a two-regime HMM are plausible candidates. It is not possible in practice to know which of the cases (a)–(c) with $d=2$, holds based on the sole ARMA(1,1) representation for ε_t^2. It is therefore of interest to consider representations for higher powers, i.e. $m>1$. First note that in the general case, the orders of the ARMA representations increase linearly with m and d. This is also true when the regimes are ARCH, but in this case the orders of the ARMA

Table 3
ARMA representations for ε_t^{2m}

	Orders	AR polynomial	MA polynomial	Intercept
Design 1: standard GARCH(1,1)				
$m=1$	(1,1)	$1-0.6L$	$1-0.427L$	1
$m=2$	(2,2)	$(1-0.6L)(1-0.4197L)$	$(1-0.5370L)(1-0.3058L)$	4.8
Design 2: independent-switching GARCH(1,1)				
$m=1$	(1,1)	$1-0.6L$	$1-0.427L$	1
$m=2$	(2,2)	$(1-0.6L)(1-0.74L)$	$(1-0.6874L)(1-0.5349L)$	4.8
Design 3: hidden-Markov model				
$m=1$	(1,1)	$1-0.6L$	$1-0.427L$	1
$m=2$	(1,1)	$1-0.6L$	$1-0.553L$	127.0
Design 4: Markov-switching ARCH(1)				
$m=1$	(3,2)	$(1-0.85L)(1-0.7215L)(1-0.1885L)$	$(1-0.8857L)(1-0.1174L)$	0.115
$m=2$		The ARMA representation does not exist (no moments of order 4)		
Design 5: Markov-switching GARCH(1,1)				
$m=1$	(3,3)	$(1-0.85L)(1-0.7215L)(1-0.1885L)$	$(1-0.8176L)(1-0.4362L)(1-0.1852L)$	0.115
$m=2$	(5,5)	$(1-0.85L)(1-0.7384L)(1-0.7215L)$ $\times(1-0.1885L)(1-0.0566L)$	$(1-0.3382L)(1-0.1879L)(1-0.0553L)$ $\times\{1-(0.8501+0.0205i)L\}\{1-(0.8501-0.0205i)L\}$	0.873
Design 6: Markov-switching GARCH(1,0)				
$m=1$	(3,3)	$(1-0.39L)(1-0.5127L)(1-0.3423L)$	$(1-0.2966L)$ $\times\{1-(0.3793+0.1856i)L\}\{1-(0.3793-0.1856i)L\}$	0.55
$m=2$	(5,5)	$(1-0.39L)(1-0.5127L)(1-0.3423L)$ $\times(1-0.3375L)(1-0.234L)$	$(1-0.496L)\{1-(0.38+0.12i)L\}\{1-(0.38-0.12i)L\}$ $\times\{1-(0.2501+0.0222i)L\}\{1-(0.2501-0.0222i)L\}$	9.48
Design 7: bilinear				
$m=1$	(0,1)	1	$1+0.268L$	1
$m=2$	(0,1)	1	$1-0.0794L$	9

Design 1: $\varepsilon_t=\sqrt{h_t}\eta_t,\quad h_t=1+0.1728\varepsilon_{t-1}^2+0.4272h_{t-1}$.
Design 2: $\varepsilon_t=\sqrt{h_t}\eta_t$, $h_t=1+(0.1415\varepsilon_{t-1}^2+1.0585h_{t-1})1_{\Delta_t=1}$, $p(1,1)=p(1,2)=p(2,1)=p(2,2)=0.5$.
Design 3: $\varepsilon_t=\sqrt{h_t}\eta_t$, $h_t=0.211_{\Delta_t=1}+46.01_{\Delta_t=2}$, $p(1,1)=0.98$, $p(2,1)=0.38$.
Design 4: $\varepsilon_t=\sqrt{h_t}\eta_t$, $h_t=(1+0.2\varepsilon_{t-1}^2)1_{\Delta_t=1}+(2+0.8\varepsilon_{t-1}^2)1_{\Delta_t=2}$, $p(1,1)=0.95$, $p(2,1)=0.1$.
Design 5: $\varepsilon_t=\sqrt{h_t}\eta_t$, $h_t=(1+0.1\varepsilon_{t-1}^2+0.1h_{t-1})1_{\Delta_t=1}+(2+0.3\varepsilon_{t-1}^2+0.5h_{t-1})1_{\Delta_t=2}$, $p(1,1)=0.95$, $p(2,1)=0.1$.
Design 6: $\varepsilon_t=\sqrt{h_t}\eta_t$, $h_t=(1+0.5h_{t-1})1_{\Delta_t=1}+(20+0.9h_{t-1})1_{\Delta_t=2}$, $p(1,1)=0.99$, $p(2,1)=0.6$.
Design 7: $\varepsilon_t=\sqrt{\eta_{t-1}^2}\eta_t$, $\eta_t\sim\mathcal{N}(0,1)$.

representations are asymmetric. By contrast, the increase of the orders in case (b) is independent of d. This is another difference between cases (a) and (b) (absence or independence of the latent process (Δ_t)) and the other cases. The HMM has the very specific feature that its ARMA orders are independent of m. It seems more difficult to discriminate between cases (a) and (b) because they provide the same ARMA orders, (m,m), whatever the power. Their differences appear by considering the coefficients of the ARMA representations.

This is illustrated in Table 3 for specific models. Design 1 concerns a standard GARCH(1,1) process. In Design 2, we choose an independent-switching regime process admitting exactly the same ARMA (1,1) representation for ε_t^2 as in Design 1. However, it is seen that the ARMA (2,2) representations for ε_t^4 differ, both by their MA and AR parts. In other words, no standard GARCH(p,q) process is compatible with the two ARMA representations of Design 2. Design 3 consists of an HMM with two regimes, providing the same ARMA representation for $m=1$ as the first two designs. Contrary to the first two designs, when m increases the orders of the ARMA representations are unchanged, as well as the AR part. Only the MA part and the intercept vary with m. In Design 4, we have a Markov-switching ARCH(1) process. We know from Table 2 that the ARMA orders of ε_t^2 are at most (2,1), and this example shows that these maximal orders are reached. For $m=2$ the ARMA representation does not exist in this case. Design 5 considers a two-regime Markov-switching GARCH(1,1) process. Again, it is worth noting that the maximal orders provided by the theory are reached: there is no common root in the AR and MA polynomials. We obtain a canonical ARMA(3,3)

for $m = 1$, and an ARMA(5,5) for $m = 2$. This is the general situation: only a very particular choice of the coefficients would provide lower orders. Now observe this: at first sight a standard GARCH(3,3) could be compatible with the ARMA(3,3) representation obtained for ε_t^2. This is not true for the following reason. The AR polynomial of ε_t^2 writes: $(1-0.85L)(1-0.7215L)(1-0.1885L) = 1-1.76L+0.9L^2-0.12L^3$, whereas a standard GARCH model would not yield a positive coefficient for L^2. The GARCH (1,0) model considered in Design 6 shows that, contrary to the GARCH(0,1) case (e.g. in Design 4), there is no order reduction when the ARCH coefficients are cancelled. The orders of the ARMA representations are the same as in Design 5. Note that, for $m = 2$, complex roots are obtained in the MA part which is impossible for standard GARCH models. Finally, Design 7 consists of a nonlinear model which does not belong to the class of this paper. Its distinguishing feature is to provide MA(1) representations for the even powers of ε_t. This is incompatible with any MS-GARCH model.

5.2. Estimating Markov-switching GARCH processes using GMM estimators

The estimation of Markov-switching GARCH(p, q) models is a difficult task when p is not equal to zero, i.e. when the regimes are not ARCH (see e.g. Klaassen, 2002). Our goal in this subsection is to show how the autocovariances derived in Section 4 can be used for fitting general Markov switching GARCH models to econometric series.

We will apply the GMM, using the autocovariances of the powers of the squared process. Assume that the orders p, q and the number of regimes d are known. The unknown parameters (the volatility coefficients $\omega(\cdot)$, $\alpha_i(\cdot)$ and $\beta_i(\cdot)$, and the non-redundant transition probabilities) are collected in a $d(d+p+q)$-dimensional vector θ. Let θ_0 be the true parameter value. We use the orthogonality conditions given by the vector equation

$$E_{\theta_0}\{h(\theta_0, \varepsilon_t, \ldots, \varepsilon_{t-K})\} = 0,$$

where the components of $h(\theta, \varepsilon_t, \ldots, \varepsilon_{t-K})$ are $\varepsilon_t^2 - E_\theta(\varepsilon_t^2)$ and $\varepsilon_t^{2m}\varepsilon_{t-\ell}^{2m} - E_\theta(\varepsilon_t^{2m}\varepsilon_{t-\ell}^{2m})$ for $\ell = 0, \ldots, k_m; m = 1, \ldots, M$ and $K = \max_m\{k_m\}$. The moments $E_\theta(\varepsilon_t^2)$ and $E_\theta(\varepsilon_t^{2m}\varepsilon_{t-\ell}^{2m})$ can be computed by means of the algorithm of Section 4.2. Let $\varepsilon_1, \ldots, \varepsilon_n$ be a realization of length $n \geqslant K$ of (ε_t). A GMM estimator $\hat{\theta}_n$ is any measurable solution of $\hat{\theta}_n = \arg\min_{\theta\in\Theta} Q_n(\theta)$, where

$$Q_n(\theta) = \mathbf{h}_n'(\theta)\mathbf{W}_n\mathbf{h}_n(\theta), \quad \mathbf{h}_n(\theta) = \frac{1}{n}\sum_{t=K+1}^{n} h(\theta, \varepsilon_t, \ldots, \varepsilon_{t-K}), \tag{12}$$

$(\mathbf{W}_n)$ is a sequence of positive definite weighting matrices converging to some symmetric and invertible non-random matrix $\mathbf{W}$, and Θ is the parameter space. If $\rho[\mathbb{P}(E\{\alpha(\cdot)\eta_t^2 + \beta(\cdot)\}^{2M})] < 1$ for each $\theta \in \Theta$, then $\{h(\theta, \varepsilon_t, \ldots, \varepsilon_{t-K})\}_t$ is an ergodic and stationary process satisfying $\|E_{\theta_0}h(\theta, \varepsilon_t, \ldots, \varepsilon_{t-K})\| < \infty$, $\forall\theta \in \Theta$. Therefore,

$$\mathbf{h}_n(\theta) \to E_{\theta_0}\{h(\theta, \varepsilon_t, \ldots, \varepsilon_{t-K})\}$$

almost surely, as $n \to \infty$. It is now clear that the consistency requires an identifiability condition of the form

$$E_{\theta_0}\{h(\theta, \varepsilon_t, \ldots, \varepsilon_{t-K})\} \neq 0 \quad \text{for all } \theta \in \Theta\backslash\{\theta_0\}. \tag{13}$$

Note that (13) entails restrictions on Θ. For instance, these restrictions should guarantee that a parameter vector obtained by inverting two regimes of θ_0 does not belong to Θ. These restrictions are standard for MS models (see e.g. Francq and Zakoïan, 2001). Moreover, it is easily seen that to achieve the identifiability condition (13) several powers m are, in general, required. Indeed, the examples of Section 5.1 reveal that different MS-GARCH models may provide the same ARMA representations for ε_t^2. Our numerical experiments showed that the identifiability problem is solved by considering representations for higher powers of ε_t^2 (see Section 6).

Now, under additional moment conditions and when θ_0 belongs to the interior of Θ, we conjecture that $n^{1/2}(\hat{\theta}_n - \theta_0)$ converges in law to a centered gaussian distribution with variance

$$\mathbf{\Sigma}(\mathbf{W}) = \mathbf{J}^{-1}\mathbf{G}\mathbf{W}\mathbf{V}\mathbf{W}\mathbf{G}'\mathbf{J}^{-1},$$

where

$$\mathbf{G} = E_{\theta_0}\frac{\partial}{\partial\theta}h'(\theta_0, \varepsilon_t, \ldots, \varepsilon_{t-K}), \quad \mathbf{J} = \mathbf{G}\mathbf{W}\mathbf{G}'$$

and

$$\mathbf{V} = \lim_{n\to\infty} \mathrm{Var}\left\{ n^{-1/2} \sum_{t=1}^{n} h(\theta_0, \varepsilon_t, \ldots, \varepsilon_{t-K}) \right\},$$

using extensions of the results in Hansen (1982). These results do not directly apply here because several assumptions (in particular, his assumptions 3.4 and 3.5) seem difficult to verify. However, the arguments given by Hansen to obtain the optimal weighting matrix apply: for any $\mathbf{W}$, the matrix $\mathbf{\Sigma}(\mathbf{W}) - \mathbf{\Sigma}(\mathbf{V}^{-1})$ is semi-positive definite, provided $\mathbf{V}$ is invertible. Therefore the optimal GMM estimator is obtained by taking for the weighting matrix $\mathbf{W}_n$ a consistent estimator of $\mathbf{V}^{-1}$. It should be noted that the invertibility of the matrix $\mathbf{J}$, or equivalently the assumption that $\mathbf{G}$ is full rank is far from being obvious, and is related to identifiability issues that will be discussed in the next section. Moreover the matrix $\mathbf{\Sigma}(\mathbf{W})$ cannot be easily estimated. Hand-coding of the derivatives involved in $\mathbf{G}$ would require much more complicated algorithms than those of Section 4.2. An alternative would be to employ automatic differentiation techniques (see e.g. Griewank and Corliss, 1991). This is left for future research. For this reason we are unable to provide standard deviations for the GMM estimators. An important diagnostic for GMM is the overidentification test proposed by Hansen (1982). For correctly specified models, the test statistic generally has a chi-square distribution with degrees of freedom equal to the number of overidentifying restrictions. However, in our model the number of overidentifying restrictions is very difficult to determine. As we will see in the next section, non-identification may occur even with a number of moments greater than the number of unknown parameters.

Details on the implementation of this GMM estimator are given in the next section.

6. Numerical illustrations

In this section we describe two sets of numerical experiments. The first subsection is devoted to identifiability issues. Given a particular model, we try to find how many moment conditions are needed to achieve consistency of GMM estimators. At this stage, we do not use Monte Carlo simulations, but we optimize numerically the asymptotic GMM objective function. In the second subsection, simulations aimed to evaluate the finite sample performance of the GMM estimator in our framework will be presented.

6.1. Identifiability issues in the GMM

We first consider the following two-regime GARCH(1,1):

$$\begin{cases} \varepsilon_t = \sqrt{h_t}\eta_t, \\ h_t = \{0.1 + 0.1\varepsilon_{t-1}^2 + 0.1h_{t-1}\}1_{\{\Delta_t=1\}} + \{0.2 + 0.1\varepsilon_{t-1}^2 + 0.7h_{t-1}\}1_{\{\Delta_t=2\}}, \end{cases} \tag{14}$$

where ε_t is iid $\mathcal{N}(0,1)$, $p(1,1) = 0.95$, $p(2,1) = 0.05$. The true parameters are collected in the vector θ_0 displayed in Table 4. To know whether the identifiability condition (13) is satisfied or not, we made the following experiments. First, we considered the GMM procedure built with the expectations of $\varepsilon_t^2, \varepsilon_t^4, \varepsilon_t^2\varepsilon_{t-1}^2, \ldots, \varepsilon_t^2\varepsilon_{t-6}^2$. Hence $K = 6$, and the components of $E_{\theta_0}h(\theta, \varepsilon_t, \ldots, \varepsilon_{t-6})$ are equal to $E_{\theta_0}(\varepsilon_t^2) - E_\theta(\varepsilon_t^2)$ and $E_{\theta_0}\varepsilon_t^2\varepsilon_{t-\ell}^2 - E_\theta(\varepsilon_t^{2m}\varepsilon_{t-\ell}^{2m})$ for $\ell = 0, \ldots, 6$. All these expectations are computed by means of the algorithm given in Section 4.2. Because ε_t^2 is an ARMA(3,3), these eight moments are sufficient to determine the complete autocovariance structure of ε_t^2. In other words, the knowledge of these eight moments is equivalent to the knowledge of $E\varepsilon_t^2$ and $E\varepsilon_t^2\varepsilon_{t-\ell}^2$ for all integers ℓ. We consider the following optimization problem:

$$\theta^* = \arg\min_{\theta\in\Theta} \|E_{\theta_0}h(\theta, \varepsilon_t, \ldots, \varepsilon_{t-K})\|. \tag{15}$$

The parameter space Θ is constrained by moment conditions (namely, in this case, the existence of $E_\theta\varepsilon_t^4$) and by the additional condition $\omega(1) < \omega(2)$. The latter constraint is made to avoid rewriting θ_0 in a trivial equivalent form by reversing the two regimes. It is clear that $\theta^* = \theta_0$ is always solution of (15). This is the unique solution of the problem if and only if θ_0 is identifiable by the GMM procedure. To solve (15), we use an optimization routine (described in the next section) which requires a starting value $\theta^{(0)}$ for θ. Choosing $\theta^{(0)} = (n^{-1}\sum\varepsilon_t^2/2, 0, 0, 2n^{-1}\sum\varepsilon_t^2, 0, 0, \frac{1}{2}, \frac{1}{2})'$ as starting value, we obtained the parameter θ^* given in Table 4. Other values of θ^* were obtained for other starting

Table 4
Illustration of the identifiability issue for model (14)

θ	$\omega(1)$	$\alpha(1)$	$\beta(1)$	$\omega(2)$	$\alpha(2)$	$\beta(2)$	$p(1,1)$	$p(2,1)$
θ_0	0.1	0.1	0.1	0.2	0.1	0.7	0.95	0.05
θ^*	0.047	0.224	0.161	0.188	0.089	0.715	0.941	0.049

Autocovariance structure of ε_t^2

	$E_\theta\varepsilon_t^2$	$E_\theta\varepsilon_t^4$	$E_\theta\varepsilon_t^2\varepsilon_{t-1}^2$	$E_\theta\varepsilon_t^2\varepsilon_{t-2}^2$	$E_\theta\varepsilon_t^2\varepsilon_{t-3}^2$	$E_\theta\varepsilon_t^2\varepsilon_{t-4}^2$	$E_\theta\varepsilon_t^2\varepsilon_{t-5}^2$	$E_\theta\varepsilon_t^2\varepsilon_{t-6}^2$
θ_0	0.49484	1.24886	0.48875	0.45411	0.42694	0.40427	0.38500	0.36846
θ^*	0.49484	1.24887	0.48875	0.45414	0.42689	0.40428	0.38504	0.36844

Autocovariance structure of ε_t^4 ($E_\theta\varepsilon_t^4\varepsilon_{t-\ell}^4$)

ℓ	0	1	2	3	4	5	6	7	8	9	10
θ_0	53.16	8.68	7.09	5.96	5.11	4.47	3.96	3.57	3.26	3.01	2.80
θ^*	47.41	7.70	6.16	5.20	4.51	3.99	3.58	3.26	3.00	2.79	2.61

The autocovariance structures of ε_t^2 for θ_0 and θ^* are the same. Those of ε_t^4 are different.

values $\theta^{(0)}$. The optimization routine reached a global minimum because $\|E_{\theta_0}h(\theta^*, \varepsilon_t, \ldots, \varepsilon_{t-K})\| \leqslant 10^{-8}$ which, in view of the tolerance chosen for the optimization routine, can be considered as being numerically equal to zero. Table 4 shows that the eight moments involved in the GMM procedure are the same (at least up to the fourth decimal) for model parameters θ_0 and θ^*. Because in both cases ε_t^2 is an ARMA(3,3), we have $E_{\theta_0}\varepsilon_t^2\varepsilon_{t-\ell}^2 = E_{\theta^*}\varepsilon_t^2\varepsilon_{t-\ell}^2$ for all $\ell \in \mathbb{N}$. Since $\theta^* \neq \theta_0$, we deduce that a GMM estimator based on any set of expectations of ε_t^2 and $\varepsilon_t^2\varepsilon_{t-\ell}^2$ cannot be consistent. Table 4 also shows that the moments $E_{\theta_0}\varepsilon_t^4\varepsilon_{t-\ell}^4$ and $E_{\theta^*}\varepsilon_t^4\varepsilon_{t-\ell}^4$ are quite different for $\ell = 1, \ldots, 10$. To make sure that the identification problem can be solved by using $m = 2$ instead of $m = 1$, the optimization in (15) was repeated with h based on the expectations of the 19 random variables $\varepsilon_t^2, \varepsilon_t^4, \varepsilon_t^2\varepsilon_{t-1}^2, \ldots, \varepsilon_t^2\varepsilon_{t-6}^2, \varepsilon_t^8, \varepsilon_t^4\varepsilon_{t-1}^4, \ldots, \varepsilon_t^4\varepsilon_{t-10}^4$. For all the starting values $\theta^{(0)}$ we tried, we found that $\theta^* = \theta_0$. This leads us to think that, based on the moments of these 19 variables, the GMM estimator should be consistent. The same experiments were repeated for other values θ_0, leading to the same conclusion: the parameter is not identifiable by the GMM based on the sole autocovariance structure of ε_t^2, but it is identifiable by also taking into account the autocovariance structure of ε_t^4.

6.2. Monte Carlo experiments

We simulated 100 independent trajectories of model (14) with sample sizes $n = 500$, 5000 and 50 000. The GMM criterion was constructed on the basis of the expectations of the 19 variables considered in the previous section. A first-step diagonal weighting matrix $\mathbf{W}_n$ was constructed by putting on the diagonal the inverses of the empirical variances of these variables.

To minimize the GMM criterion, we used several algorithms that do not require calculating derivatives. These algorithms were used jointly and are described in the book by Press et al. (1990) (see the subroutines AMOEBA and POWELL). To protect against the possibility of being trapped in local minima, we used a large number of starting values. These values were randomly selected in the parameter space. The existence of the relevant moments was verified, for each proposed value. Optimized values were then found with each of these starting values and the one that gave the smallest value for the objective function was in turn used as a starting value. The same algorithms were implemented on this value but with more restrictive tolerance criteria than the first time, thus producing a first-step estimator $\hat{\theta}_n$. A second-step weighting matrix based on $\hat{\theta}_n$ was then constructed. To estimate the inverse of the optimal matrix $\mathbf{V}^{-1}$ of Section 5.2, we used the Newey–West (1987) estimator. This estimator is defined as

$$\hat{\mathbf{V}} = \frac{1}{n}\sum_{t=K+1}^{n} \bar{\mathbf{h}}_t\bar{\mathbf{h}}_t' + 2\sum_{i=1}^{s}\left(1 - \frac{i}{s+1}\right)\frac{1}{n}\sum_{t=K+1}^{n} \bar{\mathbf{h}}_t\bar{\mathbf{h}}_{t-i}',$$

where

$$\bar{\mathbf{h}}_t = h(\hat{\theta}_n, \varepsilon_t, \ldots, \varepsilon_{t-K}) - \frac{1}{n}\sum_{s=K+1}^{n} h(\hat{\theta}_n, \varepsilon_s, \ldots, \varepsilon_{s-K})$$

Table 5
Characteristics of the empirical distribution of 100 GMM estimates of the Markov-switching GARCH(1, 1) model

n		Value	Mean	RMSE	Min.	Q_1	Q_2	Q_3	Max.
500	$\omega(1)$	0.10	0.1513	0.2453	0.0000	0.0000	0.0369	0.1377	0.8784
	$\omega(2)$	0.20	0.2541	0.3088	0.0000	0.0466	0.1867	0.3855	2.5454
	$\alpha(1)$	0.10	0.1055	0.0796	0.0000	0.0310	0.1081	0.1782	0.2713
	$\alpha(2)$	0.10	0.0785	0.1820	0.0000	0.0115	0.0549	0.0955	1.8012
	$\beta(1)$	0.10	0.1038	0.1358	0.0000	0.0067	0.0499	0.1740	0.7547
	$\beta(2)$	0.70	0.4804	0.5597	0.0000	0.2203	0.4363	0.6296	4.8587
	$p(1, 1)$	0.95	0.9549	0.1122	0.2626	0.9562	0.9986	1.0000	1.0000
	$p(2, 1)$	0.05	0.0263	0.1033	0.0000	0.0000	0.0021	0.0292	0.9999
5000	$\omega(1)$	0.10	0.0599	0.1504	0.0000	0.0000	0.0001	0.0474	0.7317
	$\omega(2)$	0.20	0.2194	0.1384	0.0000	0.1376	0.1911	0.2645	0.7924
	$\alpha(1)$	0.10	0.1215	0.0804	0.0000	0.0537	0.1425	0.1847	0.2558
	$\alpha(2)$	0.10	0.0850	0.0340	0.0126	0.0644	0.0834	0.1057	0.1753
	$\beta(1)$	0.10	0.2017	0.1963	0.0000	0.0471	0.1987	0.2997	0.7533
	$\beta(2)$	0.70	0.6251	0.1801	0.0000	0.5452	0.6651	0.7381	0.8793
	$p(1, 1)$	0.95	0.9484	0.0243	0.8660	0.9335	0.9514	0.9647	1.0000
	$p(2, 1)$	0.05	0.0331	0.0243	0.0000	0.0222	0.0310	0.0406	0.1199
50 000	$\omega(1)$	0.10	0.0411	0.0712	0.0000	0.0001	0.0344	0.0687	0.1648
	$\omega(2)$	0.20	0.1925	0.0578	0.1194	0.1515	0.1753	0.2189	0.4510
	$\alpha(1)$	0.10	0.1630	0.0934	0.0000	0.1417	0.1832	0.2029	0.2824
	$\alpha(2)$	0.10	0.0911	0.0142	0.0521	0.0838	0.0909	0.0977	0.1154
	$\beta(1)$	0.10	0.1441	0.1169	0.0001	0.0610	0.1182	0.1991	0.5427
	$\beta(2)$	0.70	0.7019	0.0594	0.5062	0.6666	0.7050	0.7451	0.8090
	$p(1, 1)$	0.95	0.9367	0.0244	0.8949	0.9216	0.9391	0.9534	0.9796
	$p(2, 1)$	0.05	0.0440	0.0100	0.0270	0.0382	0.0435	0.0479	0.0675

and s is a lag truncation parameter. In the simulations reported in Table 5, we used $s = [n^{1/5} + 1]$. Using this new weighting matrix, a second-step GMM estimator was obtained. The previous step was repeated several times until stabilization.

The simulation results presented in Table 5 correspond to model (14). As can be seen, the results are in general quite satisfactory. In particular the transition probabilities as well as the parameters of the second regime appear particularly well estimated even for moderate sample sizes. The parameters of the first regime seem more difficult to estimate. This is not very surprising since this regime is not far from a white noise. We tried other values for the lag truncation parameter s, and we took into account additional moment conditions in the GMM procedure, but without significant changes in the results.

7. Standard & Poor 500 stock price index

We now turn to an example with real data, namely the daily log returns of the Standard & Poor 500. We have a sample of daily prices from January, 2, 1979 through December, 31, 2001. Data for the week of the October 1987 crash have been suppressed from the stock-index data. Fig. 1 displays a plot of the series on the whole period.

Daily log returns are $\log(p_t/p_{t-1})$, where p_t is the closing price of the S&P 500 Index on trading day t. The logarithmic transformations of index returns are often thought to approximately follow random walks with heteroskedastic disturbances. We denote by $(r_t)_{t=1,\ldots,5798}$, the mean-corrected first differences of the logs of the S&P index. Fig. 1, as well as some standard preliminary tests, is consistent with martingale difference log returns r_t.

These data show many of the stylized facts that would lead to consideration of standard GARCH models: lack of serial correlation, heavy-tailed marginal distribution and volatility clustering. To see if there is any statistical evidence that standard GARCH models should be rejected in favor of a MS-GARCH model, we investigate the second-order properties of (r_t^2) and (r_t^4).

Based on AIC and BIC information criteria, on the t-ratios, and on the inspection of the residuals, an ARMA(3,3) was selected for the squared centered-log returns r_t^2, and an AR(5) representation was selected for r_t^4. The fitted models are

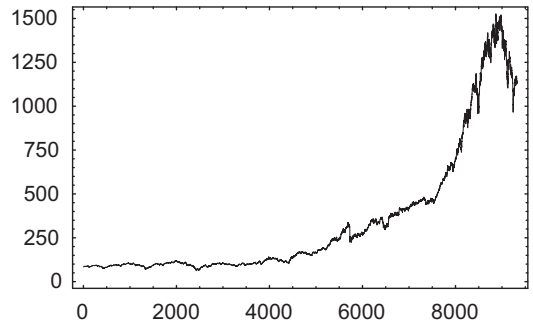


Fig. 1. Daily S&P 500 index (p_t).

Table 6
ARMA models for powers of the S&P 500 returns

ARMA(3,3) for (r_t^2)								
Parameter	$E\varepsilon_t^2$	$a_1^{(1)}$	$a_2^{(1)}$	$a_3^{(1)}$	$b_1^{(1)}$	$b_2^{(1)}$	$b_3^{(1)}$	$\mathrm{Var}(u_t^{(1)})$
Estimate	0.924	0.367	−0.330	−0.951	0.415	−0.237	−0.889	2.217
Std dev.	0.097	0.011	0.011	0.010	0.014	0.015	0.013	–
AR(5) for (r_t^4)								
Parameter	$E\varepsilon_t^4$	$a_1^{(2)}$	$a_2^{(2)}$	$a_3^{(2)}$	$a_4^{(2)}$	$a_5^{(2)}$	$\mathrm{Var}(u_t^{(2)})$	
Estimate	6.157	−0.077	−0.020	−0.017	−0.011	−0.066	69.900	
Std dev.	1.136	0.013	0.013	0.013	0.013	0.013	–	

The notations are those of Eq. (11).

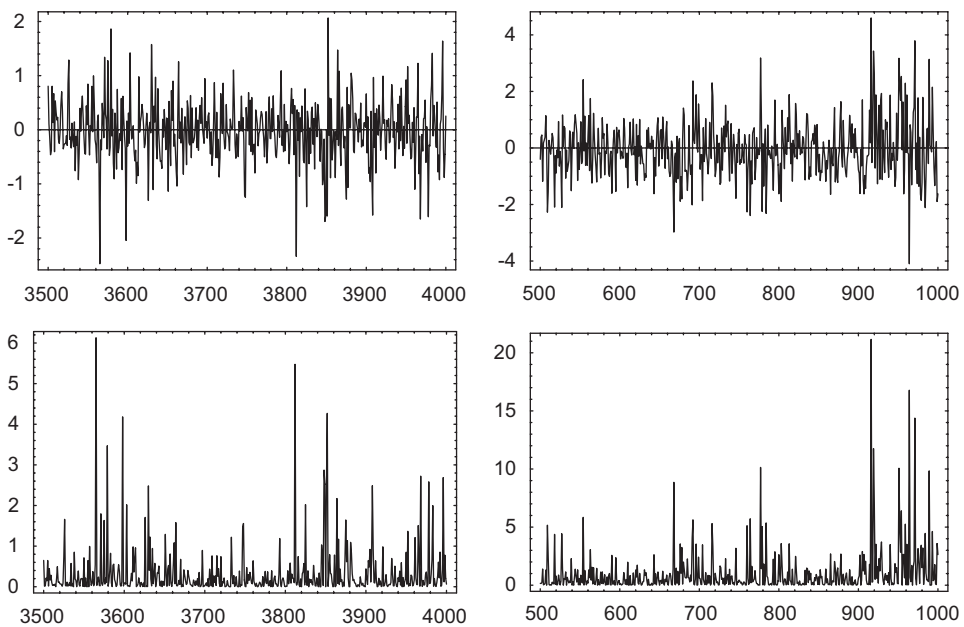


Fig. 2. Top panels: samples of size 500 of the daily S&P 500 log returns r_t; bottom panels: squares of the log returns.

given in Table 6. To obtain those results we used a standard routine, namely the ARIMA procedure of SAS, which relies on the assumption of an iid noise. Caution is therefore required in analyzing the statistical significance of the estimates. Alternatively, one could take into account the possible noise dependence structure in the estimation of the asymptotic

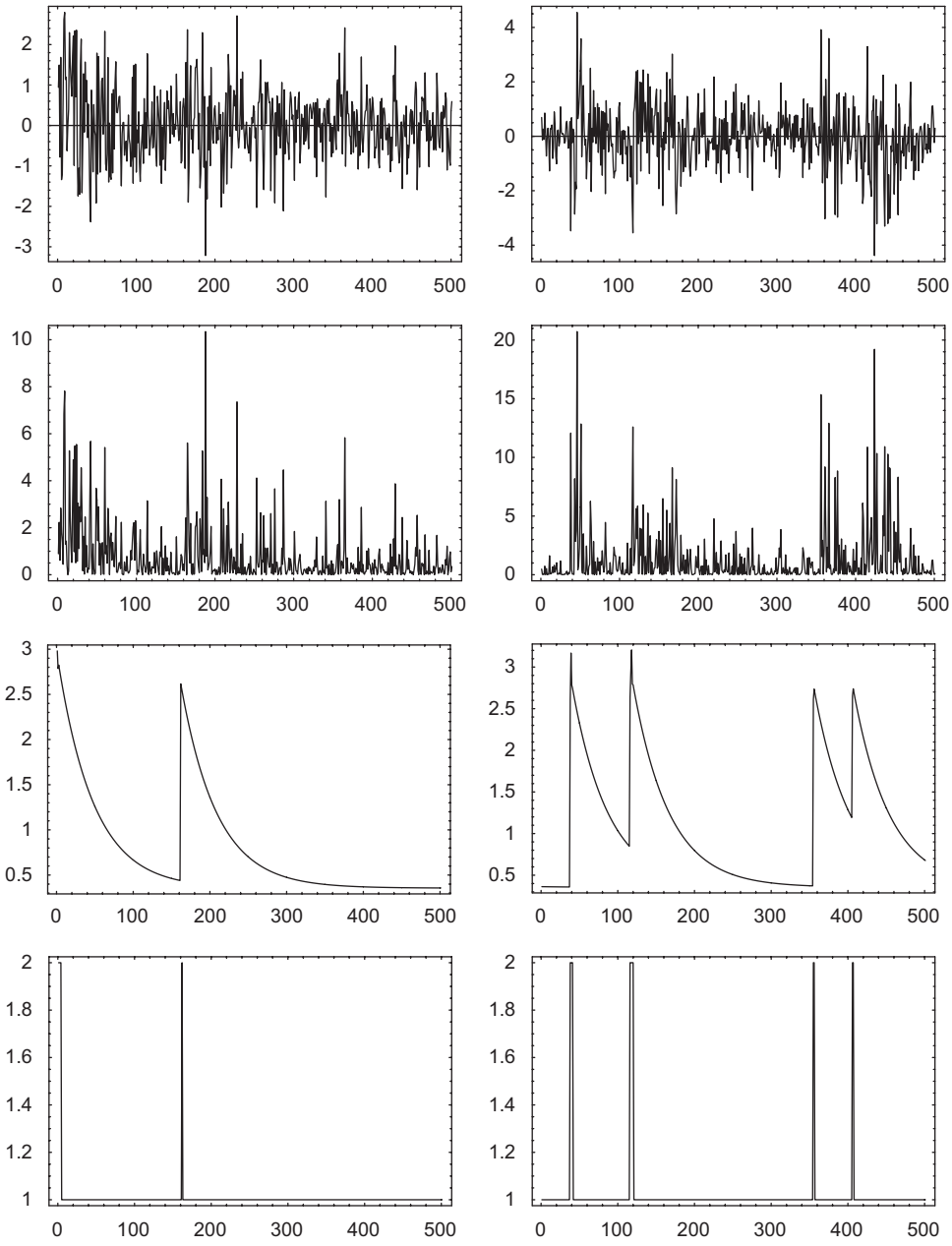


Fig. 3. Top panels: simulated trajectories of size 500 of model (16) (the 2-regime GARCH(1,1) fitted to the S&P series); second panels: squares of the simulated trajectories; third panels: volatility process (h_t); bottom panels: regime (Δ_t).

covariance matrix of the estimators (see e.g. Francq and Zakoïan, 2000). For simplicity, and since the estimation of the ARMA coefficients is not our primary concern in this paper, we did not follow this approach.

Despite these reserves, it seems that the ARMA representations are hardly compatible with a standard GARCH model. From Table 2, a standard GARCH(1,1) model would entail an ARMA(1,1) representation for r_t^2 and an ARMA (2,2) representation for r_t^4. This is not what is obtained in Table 6. The orders of the ARMA model for r_t^2 are of course compatible with a standard GARCH(3,3) model but the signs of the estimated coefficients of r_{t-1}^2 and $u_{t-1}^{(1)}$ violate the positivity constraints.

The autocovariance structures of ε_t^2 and ε_t^4 suggest considering MS-GARCH(1,1) models. The following two-regime model was fitted:

$$h_t = \begin{cases} 0.0075 + 0.0000 r_{t-1}^2 + 0.9789 h_{t-1} & \text{when } \Delta_t = 1, \\ 2.5851 + 0.0360 r_{t-1}^2 + 0.0568 h_{t-1} & \text{when } \Delta_t = 2 \end{cases} \tag{16}$$

with estimated transition probabilities $\hat{p}(1,1) = 0.9949$ and $\hat{p}(2,2) = 0.6116$. It is seen that the chain is much more persistent and frequent ($\hat{\pi}(1) = 0.9872$ and $\hat{\pi}(2) = 0.0128$) in the first regime than in the other one. We also find that the volatility is much more persistent in the first regime ($\hat{\alpha}(1)+\hat{\beta}(1)=0.9789$) than in the second one ($\hat{\alpha}(2)+\hat{\beta}(2)=0.0928$). The 'marginal variances' computed within the two regimes ($\hat{\omega}(1)/\{1 - \hat{\alpha}(1) - \hat{\beta}(1)\} = 0.3555$ and $\hat{\omega}(2)/\{1 - \hat{\alpha}(2) - \hat{\beta}(2)\} = 2.8495$) reveal that one role of the second regime may be to take into account large innovations. Note that the average time spent in the first regime is 196, which corresponds to approximately 9 months, whereas it is 2.5 in the second regime.

In Fig. 3, simulated trajectories of the estimated model (16) reveal a noticeable resemblance with the sample paths of the real series displayed in Fig. 2. A typical behavior of the S&P series is shown in the top-left panel of Fig. 2: around the 3560th log-return, a sudden negative jump occurs, followed by other jumps of decreasing amplitudes over a period of approximately 230 log-returns (about one year). A better view of this behavior is given by the bottom-left panel displaying the squares of the log-returns. A similar pattern can be observed in the top-two left panels of Fig. 3, for the simulated series. An important shock is observed just before the 200th simulation, followed by other jumps of decreasing amplitudes over a period of length 300. For these simulated data, we are able to show the volatility trajectory. In the third left panel, a sudden increase in volatility before 200, due to a regime change of the chain, can be seen. Then the chain switches immediately to the first regime and remains in this regime for long. Although very brief (only one period, as shown in the bottom left panel), the visit of the chain to the second regime has an extremely persistent effect: the volatility decreases very slowly because $\beta(1)$ is close to 1. Interestingly, the sudden increase of the volatility does not entail an immediate shock on the observed series (a delay of about 20 dates is observed between the regime change and the biggest shock in the left panels of Fig. 3). Even when h_t is large ε_t^2 may be small, but because the volatility remains high for long, a large shock is likely to be observed shortly after the regime change. Other sub-samples of the S&P series show different behaviors. It is of course difficult to infer on volatility based on the squares of the observed series. It seems however that the sub-sample displayed in the right panels of Fig. 2 presents similarities with that of the simulation in the right panels of Fig. 3. Both series display shocks of high magnitude (relative to the left panels). The simulated sub-sample is also characterized by more frequent regime changes than in the first simulation (4 instead of 2), and similar remarks can reasonably be made concerning the real sub-sample. Note also that the decrease of the shocks magnitude is still apparent, but over shorter sub-periods.

8. Discussion

This paper reviewed MS-GARCH models, which allow the volatility of financial time series to have different dynamics according to the state of an unobserved Markov chain. When (ε_t) is a MS-GARCH process, under appropriate moment conditions, the autocovariance structures of (ε_t^2), and also of (ε_t^{2m}) for $m \geqslant 2$, are those of ARMA models. The ARMA representations corresponding to different powers are found to characterize important subclasses of the general model. The ARMA representations are also informative on the orders and the number of regimes of the underlying MS-GARCH process. A practical way of computing the autocovariances and deriving the ARMA coefficients has been provided. A GMM procedure based on the ARMA autocovariances has been derived to estimate the MS-GARCH parameters. The behavior of the estimators has been studied via simulations, showing satisfactory results. We have also seen how to determine if a given set of autocovariance structures is rich enough to identify a unique MS-GARCH parameter. The methodology has been applied to stock index returns to show that MS-GARCH models can mimic behaviors that would be hard to capture using standard volatility models.

Acknowledgments

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Appendix A. Technical details

We will need the following lemmas which can be found in FZ, and whose proofs are obvious.

Lemma 7. *Let, for $i \geqslant 1$, X_{t-i} be an integrable variable belonging to the σ-field generated by $\{\varepsilon_{t-s}; s \geqslant i\}$. Then*

$$\pi(k)E[X_{t-i}|\Delta_t = k] = \sum_{j=1}^{d} E[X_{t-i}|\Delta_{t-i} = j]p^{(i)}(j,k)\pi(j).$$

Lemma 8. *Let (ε_t) be the non-anticipative, strictly stationary solution of model (1) and let X_{t-i} be a squared integrable variable belonging to the σ-field generated by $\{\varepsilon_{t-s}; s \geqslant i\}$. Assume that $E\varepsilon_t^{4m} < \infty$. Then, if $j < i$, we have for any $m' \in \{0, 1, \ldots m\}$*

$$E[\varepsilon_{t-j}^{2m'} X_{t-i}|\Delta_t = k] = \mu_{2m'} E[h_{t-j}^{m'} X_{t-i}|\Delta_t = k]$$

and, for non-negative integers m' and m'' such that $m' + m'' \leqslant m$

$$E[\varepsilon_{t-j}^{2m'} h_{t-j}^{m''} X_{t-i}|\Delta_t = k] = \frac{\mu_{2m'}}{\mu_{2(m'+m'')}} E[\varepsilon_{t-j}^{2(m'+m'')} X_{t-i}|\Delta_t = k].$$

Proof of Theorem 3. Note that (10) was already established in FZ (Proof of Theorem 3). For convenience we recall the proof. For $\ell > 0$, and for all $m' \in \{0, 1, \ldots, m\}$, $k \in \mathscr{E}$,

$$E[\varepsilon_t^{2m'} \varepsilon_{t-\ell}^{2m}|\Delta_t = k] = \mu_{2m'} \left\{ \sum_{m''=0}^{m'} \binom{m'}{m''} E[\{\omega(\Delta_t)\}^{m'-m''} \{\alpha(\Delta_t)\varepsilon_{t-1}^2 + \beta(\Delta_t)h_{t-1}\}^{m''} \varepsilon_{t-\ell}^{2m}|\Delta_t = k] \right\}. \tag{A.1}$$

Moreover, for $\ell > 1$

$$\begin{aligned} &E[\{\alpha(\Delta_t)\varepsilon_{t-1}^2 + \beta(\Delta_t)h_{t-1}\}^{m''} \varepsilon_{t-\ell}^{2m} \mid \Delta_t = k] \\ &\quad = E[\{\alpha(\Delta_t)\eta_{t-1}^2 + \beta(\Delta_t)\}^{m''} h_{t-1}^{m''} \varepsilon_{t-\ell}^{2m} \mid \Delta_t = k] \\ &\quad = E\{\alpha(k)\eta_{t-1}^2 + \beta(k)\}^{m''} E[h_{t-1}^{m''} \varepsilon_{t-\ell}^{2m} \mid \Delta_t = k] \\ &\quad = E\{\alpha(k)\eta_{t-1}^2 + \beta(k)\}^{m''} \frac{1}{\mu_{2m''}} E[\varepsilon_{t-1}^{2m''} \varepsilon_{t-\ell}^{2m} \mid \Delta_t = k] \end{aligned}$$

by Lemma 8. Hence, in view of (A.1) and Lemma 7, we have for $\ell > 1$

$$\begin{aligned} \pi(k)E[\varepsilon_t^{2m'} \varepsilon_{t-\ell}^{2m} \mid \Delta_t = k] &= \pi(k) \sum_{m''=0}^{m'} \xi_{m',m''}(k) E[\varepsilon_{t-1}^{2m''} \varepsilon_{t-\ell}^{2m} \mid \Delta_t = k] \\ &= \sum_{m''=0}^{m'} \sum_{j=1}^{d} \xi_{m',m''}(k) E[\varepsilon_{t-1}^{2m''} \varepsilon_{t-\ell}^{2m} \mid \Delta_{t-1} = j]p(j,k)\pi(j) \end{aligned} \tag{A.2}$$

which can be written in vector form as

$$\Pi(E[\varepsilon_t^{2m'} \varepsilon_{t-\ell}^{2m} \mid \Delta_t = \cdot]) = \sum_{m''=0}^{m'} \mathbb{P}(\xi_{m',m''}) \Pi(E[\varepsilon_t^{2m''} \varepsilon_{t-\ell+1}^{2m} \mid \Delta_t = \cdot]) \tag{A.3}$$

for all $\ell > 1$. Because (A.3) holds for all $m' \in \{0, 1, \ldots, m\}$, we have for $\ell > 1$, $W_m(\ell) = A_m W_m(\ell - 1)$ where $W_m(\ell)$ and A_m are defined in (6) and (4).

Note that (A.1)–(A.2) and (A.3) remain valid when $\varepsilon_{t-\ell}^{2m}$ and $\varepsilon_{t-\ell+1}^{2m}$ are replaced by 1. Therefore,

$$\Pi(E[\varepsilon_t^{2m'} \mid \Delta_t = \cdot]) = \sum_{m''=0}^{m'} \mathbb{P}(\xi_{m',m''})\Pi(E[\varepsilon_t^{2m''} \mid \Delta_t = \cdot]),$$

for $m' = 0, \ldots, 2m$. Note that $E\varepsilon_t^{2m'} < \infty$ is equivalent to

$$\rho[\mathbb{P}(E\{\alpha(\cdot)\eta_t^2 + \beta(\cdot)\}^{m'})] = \rho[\mathbb{P}(\xi_{m',m'})] < 1.$$

Hence $I - \mathbb{P}(\xi_{m',m'})$ is invertible and $\Pi(E[\varepsilon_t^{2m'} \mid \Delta_t = \cdot])$ is recursively computed from (8).

Now let us turn to the autocovariances. First note that, for $m' \leqslant m$

$$\gamma_{\varepsilon^{2m'}}(0) := \mathrm{Var}(\varepsilon^{2m'}) = e_d'\Pi(E[\varepsilon_t^{4m'} \mid \Delta_t = \cdot]) - \{e_d'\Pi(E[\varepsilon_t^{2m'} \mid \Delta_t = \cdot])\}^2.$$

We need, for the case $\ell = 1$, a formula analogous to (A.2). We first compute

$$\begin{aligned}
&E[\{\alpha(\Delta_t)\varepsilon_{t-1}^2 + \beta(\Delta_t)h_{t-1}\}^{m''}\varepsilon_{t-1}^{2m} \mid \Delta_t = k] \\
&\quad = E[\{\alpha(\Delta_t)\eta_{t-1}^2 + \beta(\Delta_t)\}^{m''}h_{t-1}^{m''+m}\eta_{t-1}^{2m} \mid \Delta_t = k] \\
&\quad = E[\{\alpha(k)\eta_{t-1}^2 + \beta(k)\}^{m''}\eta_{t-1}^{2m}]E[h_{t-1}^{m''+m} \mid \Delta_t = k] \\
&\quad = E[\{\alpha(k)\eta_{t-1}^2 + \beta(k)\}^{m''}\eta_{t-1}^{2m}]\frac{1}{\mu_{2(m''+m)}}E[\varepsilon_{t-1}^{m''+m} \mid \Delta_t = k].
\end{aligned}$$

Hence, in view of (A.1) and Lemma 7, we obtain

$$\begin{aligned}
&\pi(k)E[\varepsilon_t^{2m'}\varepsilon_{t-1}^{2m} \mid \Delta_t = k] \\
&\quad = \sum_{m''=0}^{m'}\sum_{j=1}^{d}\xi_{m,m',m''}^*(k)E[\varepsilon_{t-1}^{2(m''+m)} \mid \Delta_{t-1} = j]p(j,k)\pi(j),
\end{aligned}$$

which can be written in vector form as (9). Remark that the $\Pi(E[\varepsilon_t^{2(m''+m)}|\Delta_t = \cdot])$'s are sub-vectors of $W_{2m}(0)$, which can be obtained from (8). Hence we are able to compute $W_m(1)$ and

$$\gamma_{\varepsilon^{2m}}(1) := \mathrm{Cov}(\varepsilon_t^{2m}, \varepsilon_{t-1}^{2m}) = e_d'\Pi(E[\varepsilon_t^{2m}\varepsilon_{t-1}^{2m} \mid \Delta_t = \cdot]) - (E\varepsilon_t^{2m})^2.$$

The other autocovariances follow from (10) and the relation

$$\gamma_{\varepsilon^{2m}}(\ell) = e_d'\Pi(E[\varepsilon_t^{2m}\varepsilon_{t-\ell}^{2m} \mid \Delta_t = \cdot]) - E(\varepsilon_t^{2m})e_d'\Pi(E[\varepsilon_{t-\ell}^{2m} \mid \Delta_t = \cdot]) = v'W_m(\ell),$$

with $v' = (e_d', 0, \ldots, 0, -E(\varepsilon_t^{2m})e_d')$. □

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